Figure 1: Identification of the UMAP cluster with highest *(stim2a;stim2b)*$^{-/-}$ likelihood. A) Schematic showing the cells preparation and processing of the 5dpf zebrafish brain, FACS sorting, library preparation and data analysis. B) The cell neighbourhood graph from our single cell transcriptomic data in UMAP embedding. Leiden clusters (resolution=1) are indicated with cluster numbers and corresponding colors. C) PHATE embedding of the data. Neuronal origin cells from *Tg* and *(stim2a;stim2b)*$^{-/-}$zebrafish are coloured in blue (Tg) and orange (stim2ab). D) Mean (left) and standard deviation (right) of *(stim2a;stim2b)*$^{-/-}$ likelihood of the cells are shown on PHATE embedding. E) Mean *(stim2a;stim2b)*$^{-/-}$ likelihood of cells from each Leiden cluster. Clusters sorted by increasing mean likelihood.

Figure 2: Marker genes of the cell cluster with highest mean *(stim2a;stim2b)*$^{-/-}$ likelihood and cell type(s) annotation. A) Expression (tracks plot) of the marker genes of UMAP (Leiden) cluster 5 across all clusters. B) Word Cloud of cell type information of the top marker genes of UMAP (Leiden) cluster 5.

Figure 3: Identification of the subcluster of cells most affected by Stim2 deletion. A) Visualization of (*stim2a;stim2b*)$^{-/-}$ likelihood of all cells from each of the UMAP clusters on PHATE embedding. Please note that the cluster numbers here reflect the sorting performed in Fig2D. Here, Cluster 29 is the UMAP Cluster 5 from Fig-2. B) The (*stim2a;stim2b*)$^{-/-}$ likelihood scores are shown for UMAP Cluster 5 (Cluster 29 in Fig 3A). C) Different numbers of subclusters of the UMAP Cluster 5 with vertex frequency clustering. Three subclusters distinguished the cells according to (*stim2a;stim2b*)$^{-/-}$ likelihood (marked in red). D) The (*stim2a;stim2b*)$^{-/-}$ likelihood of cells from all the subclusters (at optimal level of grouping) from each of the selected 14 UMAP clusters (based on the number of member cells with high likelihood). The subclusters were sorted by mean (*stim2a;stim2b*)$^{-/-}$ likelihood of the subclusters. The three subclusters from UMAP Cluster 5: 4, 44, and 63 – represent the low-, mid-, and high-likelihood subclusters, respectively. E) The same subclusters, colored in blue, orange and green, respectively, are shown on PHATE embedding. F) Scatterplot of individual (*stim2a;stim2b*)$^{-/-}$ likelihood of cells from each subcluster. Cells are colored based on their sample of origin. TgX: *Tg* samples, stim2abX: (*stim2a;stim2b*)$^{-/-}$ samples. G) Marker plot of subclusters with low, mid, and high (*stim2a;stim2b*)$^{-/-}$ likelihood. Differentially expressed (DE) genes (qvalue < 0.0001; gene name on y-axis, left) with cell type information (y-axis, right) available in our curated list. Color of each point indicates the expression of each gene in each cluster (darker and brighter colors indicate lower and higher expression, respectively). The size of each point indicates how differentially expressed each gene is in each cluster. Subcluster 4, 44, and 63 are the low, mid, and high *(stim2a;stim2b)*$^{-/-}$ likelihood subclusters, respectively.

Figure 4: Expression changes between conditions in the full cluster vs in VFC clusters. A) Genes with differentially increased expression in subcluster 63. B) and C) Selected top genes with differentially decreased expression in subcluster 63 (except *elavl3*, which is included for comparison with *elavl4*). In each subpanel, the rows (top to bottom) show the expression distribution of the corresponding genes in subcluster 4, 44, and 63, and in the full cluster (UMAP Leiden cluster 5), respectively. Subclusters 4, 44, and 63 are the low, mid, and high *(stim2a;stim2b)*$^{-/-}$ likelihood subclusters, respectively. Distribution and mean line of gene expression in cells from control (*Tg*) and (*stim2a;stim2b*)$^{-/-}$ Zebrafish are shown in blue and red, respectively.

Supplementary Figure S1: MELD Results. A) Sample associated density estimates computed with MELD algorithm are visualized on PHATE plot. B) Sample associated relative likelihood estimates across replicates of *(stim2a;stim2b)*$^{-/-}$ Zebrafish shown on PHATE plot.

Supplementary Figure S2: VFC clustering Results. VFC clustering of all Leiden clusters at 2-5 levels of sub-clustering are shown. The chosen level of sub-clustering for each cluster is outlined with a red box.

Supplementary Figure S3: Marker plot of subclusters with low, mid, and high (*stim2a;stim2b*)$^{-/-}$ likelihood. Differentially expressed (DE) genes (qvalue < 0.0001; y-axis, left) with cell type information (y-axis, right), as well as the non-redundant set of top 50 (by log fold change) up- and down-regulated DE genes for each cluster are shown. Genes without known cell type information (in our curated list) are marked as “unknown” (y-axis, right). Color of each point indicates the expression of each gene in each cluster (darker and brighter colors indicate lower and higher expression, respectively). The size of each point indicates how differentially expressed each gene is in each cluster. Subclusters 4, 44, and 63 are the low, mid, and high *(stim2a;stim2b)*$^{-/-}$ likelihood subclusters, respectively.

Supplementary Figure S4: Phenotype ontology terms and member genes. Selected Zebrafish Phenotype Ontology terms significantly enriched in significantly downregulated genes in the high (stim2a;stim2b)$^{-/-}$ likelihood subcluster (sublcuster 63) are shown along with their member genes in this binary clustermap. If a gene is a member of the ontology term, the membership is marked in black.